

● DAY-1 OFFLINE MODIFICATION (HACKATHON-GRADE)

Theme: Risk Reasoning, Confidence Scoring & Explainable Decisions

🕒 Duration: 4 Hours

◆ Day-1 Context

Your system can detect anomalous crowd behavior.

In real deployments, wrong risk interpretation is worse than no detection:

- Too many false alarms → panic, loss of trust
- Missed high-risk events → real harm

Authorities therefore require:

1. Risk severity
2. Confidence in that risk
3. Clear justification
4. Consistency over time

💡 Your Challenge

Upgrade your system from “anomaly detection” to “decision-grade risk reasoning.”

🎯 Day-1 Core Objectives

Participants must implement a Risk Reasoning Engine that:

- Converts raw anomalies into risk scores
- Maps scores into risk levels
- Produces human-understandable explanations
- Maintains temporal consistency (no random flips)

CORE LOGIC TASKS (ALL MANDATORY)

1 Multi-Signal Risk Scoring Engine (Mandatory)

Instead of direct LOW / MEDIUM / HIGH labels, your system must:

Step 1: Compute a Risk Score (0–100)

The score must be derived from at least THREE signals:

Signal	Description
Density Change	Rate of crowd density increase
Motion Speed	Sudden acceleration or deceleration
Directional Chaos	Instability in movement vectors
Persistence	How long anomaly continues
Spread	How much area is affected

- ✓ Signals may be normalized
- ✓ Weighting must be explained
- ✓ Hard-coding scores is NOT allowed

Step 2: Risk Level Mapping

Risk Score	Risk Level
0–30	LOW
31–70	MEDIUM
71–100	HIGH

2 Temporal Stability Constraint (Mandatory)

Real systems do not change risk levels every frame.

Your system must implement temporal smoothing, such as:

- Sliding window logic
- Risk persistence rules
- Cool-down periods

Example Rule:

"HIGH risk must persist for at least 3 consecutive frames before escalation."

Explainable Decision Generator (Mandatory)

Each alert must include structured reasoning.

Required Explanation Format :

Time: 18:42:09

Risk Score: 82

Risk Level: HIGH

Primary Cause: Rapid crowd acceleration

Supporting Factors:

- Density increase over threshold
- Directional instability detected

Confidence: 0.87

- ✓ Explanation must match computed signals
- ✓ Confidence must be derived logically
- ✓ Copy-paste explanations are NOT allowed

False-Alarm Awareness Logic (Mandatory)

Your system must include at least ONE rule to suppress false positives, such as:

- Ignore short-lived spikes
- Require multiple signals to agree
- Downgrade risk when motion normalizes quickly

Judges will test edge cases.

Explicitly Forbidden

-  UI dashboards
 -  Manual labeling
 -  Identity or face logic
 -  Black-box decisions without explanation
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Technical Constraints

- CPU-only allowed
 - No large-scale retraining
 - Extend previous system only
 - Rule-based, ML-based, or hybrid allowed
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Day-1 Deliverables

-  Updated system code
-  Sample alert logs (with explanations)
-  A 1-page logic document explaining:
 - Risk score formulation
 - Weighting rationale
 - False alarm handling